# Jenkins Setup & Execution

## Jenkins Setup Details

- Location: QA Jenkins - Performance Testing

### Common Setup Folder

- Includes all necessary environment property files
- Includes common YAML file
- Includes key file for encryption

### Details

- Uses Jenkins parameters to pass into Docker execution to handle environment, execution details, etc.
  - Will load environment variables based on setup/<environment>.properties
  - Will set environment variables from Jenkins parameters
- Runs Docker instance loading the common YAML + selected YAML files
- Docker instance expects all necessary properties to be passed to JMeter as environment variables
  - JMeter script should have the associated step enabled for proper execution

### Parameters

- server
  - Execution server which must also be setup separately in Jenkins
- yaml
  - Taurus YAML file including relative path from /home/devops/ags-endurance
- environment
  - Environment script will run against including loading of associated properties from common folder
- encrypted
  - Account credentials are/are not encrypted
- concurrency
  - Number of users
- duration
  - Duration of test
- ramp
  - Time to ramp to full number of users

## Deploying to Server

### Script Setup

- [REQUIRED] Make sure the *Environment Setup - Remote* step is enabled (all other environment steps are disabled)
- [OPTIONAL/RECOMMENDED] Disable reporting steps
  - *Aggregate Report*
  - *View Results Tree*

### Deployment

- Create necessary folders based on script functionality starting from: /home/devops/ags-endurance (make sure to maintain correct folder structure as in GitLab)
  - Copy over JMX file
  - Copy over necessary CSV file(s)
  - Copy over remote YAML file
- Update common properties (if necessary)
  - /home/devops/ags-endurance/setup
- Repeat for all servers

## Adding Script to Jenkins

### Configuration (Single execution pipelines)

- Script execution by concurrency
  - Configure AGSUP_ENDURANCE pipeline
    - Add script to yaml parameter (including path relative from /home/devops/ags-endurance)

- Script execution by throughput
    - Configure AGSUP_ENDURANCE_THROUGHPUT pipeline
        - Add script to yaml parameter (including path relative from /home/devops/ags-endurance)

## Configuration (Suite pipelines)

- Script execution by throughput
    - Configure AGSUP_ENDURANCE_THROUGHPUT_SUITE pipeline
        - Add script execution by server using AGSUP_ENDURANCE_THROUGHPUT pipeline

# Debugging

## Logging & Artifacts

- Artifacts by default are not preserved
- Enable artifact saving to server
    - To enable, add folder /tmp/artifacts mapping to Docker execution call which will make sure all files are left on server after test completes
        - See pipeline description for exact command
    - Due to available server space, do not leave enabled by default
- Enable artifact upload to Blazemeter
    - To enable, set "upload-artifacts" in associated YAML to true
    - Artifacts will be accessible in the Blazemeter report Logs section

# Resource Management

## General

- Given limited server resources, each execution needs to be allocated CPU and RAM
- Docker is used to run the JMeter execution and has its own resource allocation
- JMeter resource allocation is limited to RAM only
- Adjust all resources based on script executions, instances, and general suite setup
- For each JMeter execution reporting to Blazemeter, you can see utilization in the Engine Health tab
    - If CPU and/or memory are maxing out, it is likely not enough resources have been allocated to that script execution
    - If CPU and/or memory are slowly climbing across the full test duration, the JMX script may need to be reviewed for cleanup and optimization
        - Verify there are no resource leaks (e.g. variables)
        - Verify all groovy scripts are cached
        - Verify no unnecessary test listeners or controllers
            - Transaction controllers especially when generating parent samples may add additional overhead

## Servers

- loadtestwt1: 8 CPU, 8GB RAM
- loadtestwt2: 8 CPU, 8GB RAM

## JMeter

- Taurus allocates RAM based on base YAML (by default)
    - Starting RAM attribute
        - memory-xms: 2G
    - Maximum RAM attribute
        - memory-xmx: 4G

## Docker

- Jenkins pipeline allocates CPU/RAM based on script
    - RAM attribute
        - --memory="4096m"
    - CPU attribute
        - --cpu-quota="400000" (in relation to cpu-period, if cpu-period is 100000, setting cpu-quota to 400000 means up to 4 CPU)